MIL-B-13994D(ME)
3 November 1972
SUPERSEDING
MIL-B-13994C(ME)
15 March 1968

MILITARY SPECIFICATION

BOAT, BRIDGE ERECTION, INBOARD ENGINE:

ALUMINUM, 27-FOOT

1. SCOPE

1.1 This specification covers a twin-screw, two-section, aluminum boat with inboard engines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| O-A-548 | - Antifreeze, Ethylene Glycol, Inhibited. |
| QQ-S-781 | - Strapping, Steel, Flat and Seals. |
| UU-T-81 | - Tags, Shipping and Stock. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |
| PPP-T-60 | - Tape: Packaging, Waterproof. |

Military

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment). |

FSC 1940

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| | |
|---|---|
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-L-2104 | - Lubricating Oil, Internal Combustion Engine, Tactical Service. |
| MIL-E-10062 | - Engines: Preparation for Shipment and Storage of. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-L-21260 | - Lubricating Oil, Internal-Combustion Engine, Preservative and Break-In. |

STANDARDS

Federal

| | |
|---|---|
| FED. STD. No. 595 | - Colors. |

Military

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-403 | - Preparation for and Installation of Rivets and Screws, Rocket and Missile Structure. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-781 | - Reliability Tests: Exponential Distribution. |

DRAWINGS

ME

| | |
|---|---|
| TA13213E1110 | - Boat, Bridge Erection, Inboard Engine: Aluminum, 27-Foot. |
| D13206E4468-1 thru -6 | - Packing of Bridge Erection Boat. |

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for propsal shall apply.



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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

U. S. ARMY HUMAN ENGINEERING LABORATORIES

HEL-STD S-1-63 - Maximum Acceptable Noise Level for Army Materiel Command Equipment.

(Application for copies should be addressed to the U. S. Army Human Engineering Laboratories, Aberdeen Proving Ground, Aberdeen, MD 44156.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

## 3. REQUIREMENTS

3.1 Description. The bridge erection boat shall be in accordance with Top Assembly TA13213E1110, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the boat. All dimensions shown on the end product drawings are based on temperatures of 68° F. through 70° F. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a bridge erection boat for examination and test within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will product boats that comply with the



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requirements of this specification. Examination and tests shall be as specified in section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more boats (see 6.2) for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.5 Reliability. The specified mean-time-between-failure (MTBF) shall be 100 hours when the boat is tested as specified in 4.6.2.11.

3.6 Maintainability. The boat shall operate as specified herein for 300 hours without maintenance other than the supplier's recommended schedule of adjustments as established by a maintenance schedule submitted by the supplier prior to test. All assemblies and installed attachments shall be accessible for servicing, repair, and replacement. Enclosures such as gear cases and reserviors which contain lubricant shall be equipped with, in addition to an accessible fill opening and dipstick, an accessible opening compatible with the sump pump to be furnished by the supplier. The ratio of man-hours of maintenance required to the hours of operation performed shall not exceed 0.15.

3.7 Loft. A full scale loft, based upon the drawing table of offsets, shall be used to determine hull mold lines as specified on the drawings. The loft shall be available to the Government at all times for determining conformance of templates and hull components. At the supplier's option, the loft shall be Government-loaned (see 3.15) or prepared by the supplier subject to the following conditions:

(a) The loaned loft shall be protected at all times and shall be returned to the Government without tool marks, smudges, holes, or other blemishes resulting from supplier usage.
(b) When prepared by the supplier, the loft shall be subject to the approval of the contracting officer before hull assembly tooling is fabricated.



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3.8 Performance. The boat shall be capable of obtaining a forward, horizontal, static towline pull of not less than 3,400 pounds at full throttle (maximum engine rpm). Under all test conditions specified herein, the coolant temperature after warmup shall operate in the range of 155° to 190° F., and the engine and transmission oil pressures shall operate in the ranges between 32 - 60 psi and 120 - 160 psi respectively. The hull, closed-circuit cooling systems, and fuel systems shall not leak as a result of the tests specified herein. All operable features shall function without binding or interference of movement. There shall be no evidence of vibration caused by unbalanced propellers, misalinement between transmission and propeller shaft coupling flanges, bent shafts, or untuned engines. The boat shall be capable of withstanding a full-throttle-forward to full-throttle-reverse sudden stop without malfunction or failure of components. The boat shall be capable of maneuvering a five-float reinforced M4T6 or Class 60 bridge raft, both by pushing and by towing. The Government will lend the necessary components to assemble the raft, in accordance with Technical Manual TM5-210, for test purposes (see 3.15 and 4.6.2.8). Under all test conditions specified herein, the boat shall start, run, and stop without any component loosening, failing, or permanently deforming. The boat shall operate within the noise levels established by HEL-STD S-1-63 when tested in accordance with 4.6.2.10.

3.9 Electromagnetic interference. The electromagnetic interference emission characteristics of the boat shall conform to MIL-STD-461, Notice 1, class III C.

3.10 Lubrication. The boat shall be assembled, run-in, tested, and delivered with the following lubricants.

3.10.1 Grease. Grease conforming to MIL-G-10924 shall be applied where relative movement between metal surfaces would occur during operation of the boat. Electrical components shall not be lubricated. Grease cups shall be filled.

3.10.2 Oil. The boat shall operate as specified herein when using oil conforming to MIL-L-2104 or MIL-L-21260, type I or type II, grade as applicable.

3.11 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I, with 1 percent copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.



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(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared, so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12 Identification marking. The marking for the boat identification plate, as specified on the drawings, shall include a two-part serial number such as 68-1 indicating the year and hull number, starting with the number "1" for the first boat fabricated. In addition, the serial number shall be marked in other locations of the boat as follows:

(a) The serial number shall be die-stamped in 1/2-inch figures on the bow and stern sections at the centerline of the deck above frame 6.
(b) The serial number shall be stenciled in 1-1/2-inch Arabic figures on the outside of the port and starboard sides of the bow and stern sections and shall be located below the gunwale near frame 6. Stenciling shall conform to MIL-T-704.

3.13 Painting. Unless otherwise specified on the drawings, the boat shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, color No. 34087, olive drab, as specified in FED. STD. No. 595. Components and parts shall be cleaned, treated, and primed prior to assembly. Components for riveted assemblies shall be painted after riveting. Other components to be painted may receive the final coat of paint before or after assembly. Following the tests specified herein, the boat shall be washed free of water scum or other foreign matter and missing paint shall be replaced.

3.14 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.4):

| Item No. | Description | Identification | Quantity f/ea. boat |
|---|---|---|---|
| 1. | Operation and Maintenance Publications Case | FSN 7520-559-9618 | 1 |
| 2. | Preservation and Depreservation Guide for Marine Equipment | DA 3256 | 1 |



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3.15 Government-loaned property. The following property will be loaned by the Government (see 3.7, 4.6.2.8 and 6.5):

| Item No. | Description | Identification | Quantity |
|---|---|---|---|
| 1. | Loft II (full scale) | Drawing 1321E5459 (Sheets 1 through 6) (When requested by the supplier) | 1 |
| 2. | Inspection equipment for Boat, Bridge Erection, Inboard Engine: Aluminum, 27-Foot | Quality Assurance Pamphlet AMSMO-FB-P715-114 | Set |
| 3. | Five-float reinforced raft with Technical Manual | TM5-210 | 1 |

3.16 Workmanship.

3.16.1 Metal fabrication. Fixtures, jigs, and templates shall be used to insure the specified configuration, alinement, and location of components. Incised scribe lines shall not be made on cadmium-coated or zinc-coated surfaces. Aluminum material having cracks, scratches, or incised scribe lines shall not be used in fabrication of the boat. Forming of aluminum alloy material shall be accomplished without heat (cold forming). Aluminum shall not be cut by methods using heat. Shearing, chipping, and flame cutting of steel shall be done neatly and accurately. All metal edges shall be free from slivers and burrs.

3.16.2 Riveting. Riveting shall conform to MIL-STD-403. Components to be riveted shall be formed and fitted prior to riveting to assure that driven rivets will not be in tension. Rivet holes in joining members shall be alined before riveting. Rivets shall not be overdriven as evidenced by seam deformation and resulting separation of seam lap edges from the surface. Riveting adjacent to welding shall be done after welding. Rivets with heads undersized or not having complete bearing on metal, or rivets otherwise defective shall be replaced. Rivets causing leakage shall be replaced unless they can be redriven without overdriving. Rivets to be replaced shall be removed by first drilling out with a drill smaller in diameter than the rivet shank. Where an enlargement of the rivet holes occurs, the holes shall be reamed to a true circle and the next larger size rivet driven. Mislocated rivet holes may be plug welded and redrilled on all riveted components except castings. Countersunk rivets with heads below acceptable depth of countersunk member shall be replaced only when (1) rivet causes leakage, (2) adjacent rivets are also defective, or (3) more than 5 percent of countersunk rivets in the same member are also defective.



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3.16.3 Welders. Welders shall be qualified in accordance with the Standard Qualification Procedure of the American Welding Society or Welding Qualifications of the American Society of Mechanical Engineers.

3.16.4 Welding. Aluminum welding shall be done by the inert gas (argon or helium), shielded-arc method. Steel may be welded by processes other than the shielded-arc method provided the mechanical properties of the welds are equivalent to the as-welded condition of the electrodes specified on the drawings. The surfaces of parts to be welded shall be free from rust, paint, scale, or other foreign matter. Oxide film on aluminum surfaces shall be removed by a solvent or by wire brushing before welding. Welds may be ground, filed, or buffed but shall not be hammered. All welded parts shall be free from cracks or other imperfections that may reduce the effectiveness of the part.

3.16.5 Interchangeability. All like components shall be interchangeable with respect to assembly, performance, and quality.

3.16.6 Assembly. All components shall be present and clean. Assembly shall be made without distorting or forcing together any components. Threaded fasteners shall be screwed tight but shall not distort components being held. Throttle-control cables shall be adjusted to provide full fuel advance at the engine. Unless otherwise specified on the drawings, instrument lines, electrical wires, and control cables shall be routed for optimum concealment and support from structural members. The propeller struts and rudder struts shall be precisely shimmed to assure alinement of rudder shaft bearings with rudder strut pivots, alinement of propeller shaft bearings with transmission coupling flange centerline, and connection of struts and braces. With the boat afloat, the transmission coupling flanges and the propeller shaft coupling flanges shall be parallel within 0.005 inch. This condition shall be checked both before and after the performance test.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Boats which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government.



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4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Disassembly inspection. Should the preproduction boat fail a performance test, it shall be disassembled in the presence of a Government representative to the extent necessary to determine the cause. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.4 Parts and components. Parts and components shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) listed on Top Assembly TA13213E1110.

4.1.5 Inspection equipment. Special final inspection equipment to be loaned to the supplier by the Government (see 3.15) is identified in the quality assurance pamphlet listed on TA13213E1110.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction boat shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction boat shall be tested as specified in 4.6.2 through 4.6.2.10. Failure of any test shall be cause for performing the inspection specified in 4.1.3.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production boats will be selected at random by the Government from boats being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance



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of an initial production boat shall not exclude the remaining boats from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production boat to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production boat(s) and shall be cause for refusal by the Government to continue acceptance of production boats until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on boats previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all boats accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each boat shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each boat shall be tested as specified in 4.6.2 through 4.6.2.5. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The boat shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Lubricants not applied as specified.
103. Cleaning, treatment, and painting not as specified.
104. Government-furnished property not mounted as specified.
105. Assembly not as specified.
106. Metal fabrication not as specified.
107. Riveting not as specified.
108. Welders not qualified as specified.
109. Welding not as specified.

4.6.2 Tests.

4.6.2.1 Conditions. Prior to tests, the boat shall be lubricated with grease and oil as specified in 3.10. The preproduction test boat, initial



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production test boats or inspection comparison test boats shall be filled to operating level with oil conforming to MIL-L-2104, grade 30. Quality conformance test boats shall be filled to operating level with oil conforming to MIL-L-21260, type I, grade as applicable for spark ignition or compression ignition engines without supercharger or turbocharger. The boat shall be complete with all safety equipment, but shall not include batteries to be furnished with the boat or other loose components unrelated to these tests. The supplier shall furnish fuel and batteries suitable to operate the boat for the specified tests. Fresh water or an antifreeze solution of ethylene glycol conforming to O-A-548 and water compatible with temperature conditions shall be used as a coolant. Prior to tests, the break-in period, as specified by the engine manufacturer, shall be completed. The engines shall be tuned for optimum operation. At the conclusion of the test each engine crankcase and transmission of each boat tested using oil conforming to MIL-L-2104, grade 30 shall be drained and refilled to the operating level with oil as specified herein for quality conformance test boats. The engine crankcase and transmission of boats tested using oil conforming to MIL-L-21260, type I, grade as applicable shall not require draining. All tests shall be performed at a single test site which will accommodate the tests specified in 4.6.2.7.

4.6.2.1.1 Test records. A log for each boat shall be kept, recording the engine oil pressures, coolant temperatures, and rpm every 30 minutes. A daily record of weather conditions, ambient water temperature, ambient air temperature, oil pressure and rate of charge shall also be kept.

4.6.2.2 Dock. Place the bow and stern sections of the boat in the water, using the lift slings to be furnished with the boat, and connect the sections. Secure the boat to a dock or pier. Check the alinement of the transmission flange couplings and the propeller shaft couplings with a feeler gage to determine conformance to 3.16.6. Bleed the propeller shaft seals to remove entrapped air. Start the engines according to instruction marking and bring to idle speed. While the engines are warming, check engine instruments, electric bilge pump, and lights for operation. Load the boat with 4,000 pounds distributed to maintain boat trim. Check for hull leaks. After engine temperature readings are stable, operate in forward gear at 2,000 rpm for not less than 1 hour. Inspect the engine cooling systems, fuel systems, the hull seams and hull openings (shaft logs, rudder shafts, and exhaust outlets) for leakage during engine operation. During the last 5 minutes of operation, record and check the engine and transmission oil pressures and coolant temperatures for conformance to 3.8. Nonconformance to 3.8 or 3.16.6 shall constitute failure of this test. Remove the 4,000 pound weights at the conclusion of this test.

4.6.2.3 Operation. Free the boat and make a trial run with an operator and two inspectors aboard. Operate and maneuver the boat in accordance



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with 4.6.2.7 for one complete cycle. At least six rapid accelerations shall be performed by suddenly dropping the throttle from idle to wide open. Observe the reaction of the boat to throttle and rudder changes and check engine oil pressure and coolant temperatures for conformance to 3.8. At the end of the trial run, stop the port engine, disconnect the portside batteries, and restart the port engine by activating the paralleling and starter switch for the port engine. Start the starboard engine in the same manner using the opposite procedure. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.4 Static pull. Fasten the boat to a stationary object with a line connected to the stern towing bitt. The line shall include a dynamometer with a capacity of not more than 5,000 pounds, calibrated in increments of not more than 100 pounds. Operate both engines simultaneously in forward gear at varying engine speeds up to full throttle. Inability to obtain the towline pull specified in 3.8 shall constitute failure of this test.

4.6.2.5 Reverse. Fasten the boat to a stationary object with a line connected to the bow cleats. Provide a means of determining the oil temperature for each transmission. Operate the engines simultaneously in reverse gear at full speed for not less than 30 minutes. Record maximum engine oil, transmission oil pressures and coolant temperatures for both power plants. Inability to maintain temperatures or pressures within the ranges specified in 3.8 or malfunction of any component shall constitute failure of this test.

4.6.2.6 Sudden stop. With the boat free to move, run full-throttle straightaway until the forward speed of the boat is constant. After warning all passengers, shift the transmission-throttle control levers from full-speed forward to full-speed reverse as rapidly as possible. Maintain full-reverse throttle until the relative motion of the boat has stopped. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.7 75-hours endurance. The following maneuvers shall be executed at maximum rpm for 75 hours:

(a)
1. Run straight for 5 minutes.
2. Execute a sudden 90° port turn and run 1 minute.
3. Execute a sudden 90° port turn and run 5 minutes.
4. Execute a sudden 90° port turn and run 1 minute.
5. Execute a sudden 90° port turn and run 5 minutes.
6. Execute a sudden 90° starboard turn and run 1 minute.
7. Execute a sudden 90° starboard turn and run 5 minutes.
8. Execute a sudden 90° starboard turn and run 1 minute.
9. Repeat this maneuver three times.

(b)
1. Execute a sudden hard over starboard rudder after maximum rpm's have been reached on a straight run and continue in a circle for 5 minutes.



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2. Repeat the above maneuver with a hard over port rudder.

(c) 1. Run a tight figure eight pattern at maximum rpm's for 5 minutes.

(d) 1. Execute a sudden acceleration, idle to full-throttle, every 7-1/2 hours.

(e) 1. Inspect the boat at the end of each day. Failure to conform to 3.8 shall constitute failure of this test.

4.6.2.8 Raft. Assemble a five-float reinforced M4T6 or class 60 bridge raft in accordance with TM 5-210 with the components furnished by the Government (see 3.8 and 3.15). Secure the boat to the raft and maneuver at full throttle for 150 hours. Two landings per hour shall be made by placing the ramp each time where it would be possible to drive a wheeled vehicle onto the raft. The time between landings shall be spent running a straight course away from the landing site, alternately executing a 180 degrees starboard or port turn, and returning to landing site. Failure to conform to 3.8 shall constitute failure of this test.

4.6.2.9 Dock. Secure the boat to the dock and operate according to the following schedule for a total of 75 hours in forward gear:

Cycle - Operation Schedule

| RPM | Duration (hours) |
|---|---|
| 1000 | 1 |
| 1700 | 2 |
| wide open throttle | 2 |
| 1700 | 2 |
| 1000 | 1 |

Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.10 Noise level. Noise levels shall be measured as prescribed in HEL. STD. S-1-63. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.11 Reliability. Using the MTBF specified in 3.5, the initial production boat shall be tested as specified in 4.6.2.7, 4.6.2.8 and 4.6.2.9 with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan I. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected



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within 30 minutes by adjustment, repair, or replacement action using controls, on-equipment tools or parts, and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
(b) Damage to the boat(s) by continued operation.
(c) Safety hazards to personnel.

Malfunctions which would not affect mission performance will not be considered failures.

4.6.2.12 Maintenance ratio. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.6. The ratio of all the manhours of maintenance required to the hours of operation performed shall be computed. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.13 Maintenance support. During testing as specified in 4.6.2.11, maintenance support elements such as manuals (when furnished as required by the contract), lubrication charts, repair parts, and special tools shall be evaluated. Inability of the boat(s) to be maintained as specified, utilizing the support elements shall constitute failure of this test and shall require revision of the document(s). Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the mean-time-between-failure.

4.7 Inspection comparison. The Government may select boats at any time during the contract production period and subject these boats to the examination specified in 4.6.1 and the tests specified in 4.6.2 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on boats selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production boats. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison boat to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison boat(s) and shall be cause for refusal by the Government to continue acceptance of production boats until evidence has



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been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on boats previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all boats accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.1. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Examination. The preservation, packaging, packing, and marking shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

| | |
|---|---|
| 110. | Materials, methods, or containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect. |
| 111. | Disassembly not as specified for level A. |
| 112. | Bolts, nuts, screws, pins, and washers not reinstalled in mating parts and secured to prevent loss for level A. |
| 113. | Unpainted exterior metal surfaces not coated with preservative for level A. |
| 114. | The engines, components, and accessories not preserved as specified. |
| 115. | Engine crankcase and transmission not filled to operating level with preservative lubricating oil and tagged as specified. |
| 116. | Electrolyte not packaged, packed, or secured as specified for level A. |
| 117. | Interior surfaces of the oil sump pump not coated with preservative as specified for level A. |
| 118. | Hand bilge pump not placed and secured in the bracket provided. |
| 119. | Boat hook not placed and secured as specified for level A. |
| 120. | Haul chock lines not secured in place for level A. |
| 121. | Rope not coiled, secured, and stowed as specified for level A. |
| 122. | Anchor light mast and bow light mast not placed and secured as specified for level A. |



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123. Drain plugs not removed, secured, and tag attached as specified.
124. Lift slings not placed and secured as specified.
125. Spotlight not placed in box provided.
126. Ring buoy not placed and secured in the brackets provided.
127. Fire extinguisher not placed and secured in its mounting bracket.
128. Coupling handles not secured as specified for level A.
129. Covers not placed and secured in place.
130. Lashing and tiedown cables and slings not secured as specified for level A.
131. Maintenance tools not preserved and packaged as specified for level A.
132. Consolidated packaging not as specified for level A.
133. Sections not placed and secured on the cradles as specified.
134. Packing of individual sections not in accordance with referenced drawings for level A.
135. Depreservation guide not prepared and attached as specified.
136. Marking missing, illegible, incorrect, or incomplete.

## 5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging and packing, that the applied preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The preproduction pack may be accomplished utilizing either the preproduction model boat or a production boat. If the preproduction model boat is utilized, any preservation, packaging and packing shall be removed by the supplier at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model boat and production boats.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be confined to the removal of parts necessary to accomplish the preservation, packaging, and packing specified herein. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in mating parts and secured to prevent their loss.

5.2.1.2 Preservatives. Preservatives specified shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.



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5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of items, accessories, or equipment requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engines. The engines, components and accessories shall be preserved in accordance with MIL-E-10062, level A, as specified for type V classification and with the following exceptions and additional requirements.

5.2.1.4.1 Lubrication system. The preservative oil conforming to MIL-L-21260, type and grade as applicable, contained in the engine crankcase and transmission (see 4.6.2.1) shall remain in the units. Oil shall be added as required to bring the oil to the specified operating level. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location in the cockpit indicating the following: "Engine crankcase and transmission are filled to the operating level with preservative lubricating oil and is good for operation until the first required oil change."

5.2.1.4.2 Electrolyte. When furnished, the electrolyte shall be packed in wood boxes. The units per pack shall be the quantity required for each boat. The electrolyte pack shall be secured to the base of the crate containing the stern section.

5.2.1.5 Oil sump pump. Interior surfaces of the pump shall be coated with preservative oil conforming to MIL-L-21260, type I, grade 20.

5.2.1.6 Hand bilge pump. The hand bilge pump shall be placed and secured in the bracket provided.

5.2.1.7 Boat hook. The boat hook shall be placed and secured in the spring clips provided.

5.2.1.8 Haul chock lines. The haul chock lines shall be secured in place.

5.2.1.9 Rope. Each length of loose rope shall be coiled and the coil secured with evenly spaced ties of cotton tape. The coils of rope shall be placed in the bow compartment as shown on the drawings (see 5.3.1).

5.2.1.10 Anchor light mast and bow light mast. The anchor light mast and bow light mast shall be placed and secured in the spring clips provided.

5.2.1.11 Drain plugs. The drain plugs in each section of the boat shall be removed and secured to a member to prevent movement. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location in the cockpit indicating: "Drain plugs removed; install before placing boat in water."

5.2.1.12 Lift sling. The lift sling shall be placed on the brackets provided in the engine compartment and secured with tape conforming to PPP-T-60, type IV.



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5.2.1.13 Spotlight. The spotlight shall be placed in the spotlight box provided and the box door secured with tape conforming to PPP-T-60, type IV.

5.2.1.14 Ring buoy. The ring buoy shall be placed and secured in the bow section in the brackets provided.

5.2.1.15 Fire extinguisher. The portable fire extinguisher shall be placed in its mounting bracket and secured.

5.2.1.16 Coupling handles. The coupling handles shall be secured in place as shown on the drawings (see 5.3.1).

5.2.1.17 Lashing and tiedown cables and slings. The loose ends of the cleat-to-truck cables shall be secured to the cradles. The stern section aft lashing shall be secured to the cleats on the stern section of the boat with soft annealed wire.

5.2.1.18 Repair parts. Repair parts shall be preserved in accordance with the preservative application criteria and applicable method(s) of preservation of MIL-P-116.

5.2.1.19 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.2.1.20 Technical publications. Technical pulications shall be preserved in accordance with MIL-P-116, method IC-1, and placed in the case provided.

5.2.1.21 Consolidated packaging. The anchor with rope, gas gage, and oil sump pump shall be consolidated in a close-fitting box conforming to PPP-B-621, class 1, style optional, or PPP-B-601, domestic type, style optional as shown on the drawings (see 5.3.1). When repair parts and tools are shipped with the boat they shall be intermediate packaged in a separate box as specified and the box secured to the top of the box containing the other components with two straps conforming to QQ-S-781, type I, class B, size as applicable. The boxes and pushing knees shall be placed and secured in the boat, and the connecting levers secured in the retracted position as shown on the drawings (see 5.3.1).

5.2.2 Level C.

5.2.2.1 Repair parts and technical publications. Repair parts and technical publications shall be preserved as specified for level A.

5.2.2.2 Boat and boat components. The combustion chambers, valves, and cooling system of the engines shall be preserved as specified for level A. The preservative oil contained in the engine crankcase and transmission



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shall not be drained. The hand bilge pump, lift slings, spotlight, ring buoy, fire extinguisher, and section covers shall be placed and secured as specified for level A. Drain plugs in each section of the boat shall be removed and secured to a member. Two tags shall be attached in a conspicuous location in the cockpit indicating that: (1) "Engine crankcase and transmission are filled to the operating level with preservative lubricating oil and is good for operation until the first required oil change," and (2) "Drain plugs are removed; install before placing boat in water." Preservation and packaging of all other components of the boat shall be such as to provide protection against deterioration and damage from the supplier to the initial destination.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each boat section with covers secured in place shall be packed as shown on Drawings D13206E4468-1 through -6.

5.3.2 Level C. Each boat section secured on its cradle (see 6.7) shall be packed to assure carrier acceptance and undamaged delivery at lowest rating from the supplier to the initial destination.

5.4 Depreservation guide. A depreservation guide shall be prepared. Place the guide in a waterproof envelope marked "Depreservation Guide" and secure in a conspicuous location on the boat. Unless otherwise specified (see 6.2 and 6.8) DA Form 3256 - Preservation and Depreservation Guide for Marine Equipment shall be used.

5.5 Marking. In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129, and the following additional requirements:

(a) Mark each end and each side of crates containing a stern section:

"STERN SECTION ONLY"
SERIAL (See 3.12)

(b) Mark each end and each side of crates containing a bow section:

"BOW SECTION ONLY"
SERIAL (See 3.12)

Lettering shall be 3 inches high and shall be placed at the top center of ends and sides.



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6. NOTES

6.1 Intended use. The boats are intended for pushing and maneuvering components of military floating bridges.

6.2 Ordering data. Procurment documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contacting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required and number of boats to be furnished, when applicable (see 3.3).
(e) When Government will furnish the property listed in 3.14.
(f) Time frame required for submission of the preproduction pack (see 5.1).
(g) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).
(h) When other than DA Form 2258 shall be used (See 5.4).

6.3 Preproduction model. Any changes or deviations of production boats from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish boats conforming to this specification.

6.4 Government-furnished property. When specified, the contracting officer should arrange to furnish the property specified in 3.14.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.15.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack and mark the boats in accordance with this specification.

6.7 Domestic transport of boats. It is desirable to transport uncrated boats within CONUS by the most expedient and least costly methods available. In addition to the method approved by the Association of American Railroads consisting of loading one complete boat on a rail flatcar, the following method of transport has proven to be adequate:

Highway trailer. Three complete boats, all sections in cradles, may be loaded on certain types of commercial boat trailers having



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a minimum length of 40 feet. Such trailers are available from several haulaway companies. Three like sections, either sterns or bows, should be loaded on the trailer bed, and the remaining three sections loaded onto racks which may be mounted in the trailer bed. Boat sections loaded on the trailer bed should have their cradles secured directly to the bed. Boat sections loaded on the racks should have their cradles secured to the racks and thence to the trailer bed. Additional tiedowns should be applied to prevent longitudinal and lateral sway of the rack-loaded sections. Chains and load binders are suggested as the most suitable means of securing the load. This loading method has been tested and found satisfactory.

6.8 DA Form 3256. The contracting officer should arrange to furnish DA Form 3256 when requested by the supplier.

6.9 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each bridge erection boat.

6.10 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of bridge erection boats. A suggested paragraph is as follows:

"Shipment of bridge erection boats shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.11 Certain provisions of this specification are the subject of international standardization agreement ABCA-Army-STD-141. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective departmental standardization offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 1940-A024



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-13994D(ME) Boat, Bridge Erection, Inboard Engine: Aluminum, 27-Foot

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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